A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

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Preliminary Progress Review (PPR) Report

Wijekoon H.M.R.K.

B.Sc. (Hons) in Information Technology Specializing in Computer Systems & Network Engineering

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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Contents

Declaration 1
1.0 INTRODUCTION 3
1.1 Purpose 3
1.2 Abbreviations 4
1.3 Overview 4
2.0 Literature Review 5
2.1 Identification and Significance of the Problem 5
3.0 Research Question 5
4.0 Research Objectives 6
4.1 Main Objective 6
4.2 Sub Objectives 6
5.0 METHODOLOGY 7
5.1 System diagram 7
5.2 My Research Component Methodology 8
5.3 Tools, Software, and Environment 9
5.4 Data needs and collection strategy 9
5.5 Experimental design and evaluation criteria 10
5.6 Tasks, sub-tasks and milestones 11
5.7 Anticipated results 13
6.0 Current Progress 14
7.0 Anticipated Benefits 14
7.1 Benefits to Users 14
7.2 Contribution to the Body of Knowledge 14
8.0 Scope and Expected Research Outcome 15
9.0 Research Constraints 15
10.0 Project Plan 16
11.0 REFERENCE 16

1.0 INTRODUCTION

The Internet of Things (IoT) has become an essential part of modern technological systems, enabling interconnected devices to sense, process, and exchange data across diverse environments. Many of these systems rely on Low-Power and Lossy Networks (LLNs), where devices operate with limited energy, memory, and processing capabilities. To support routing in such constrained environments, the Routing Protocol for Low-Power and Lossy Networks (RPL) has been widely adopted as the standard routing protocol.

Although RPL is efficient and lightweight, it was not originally designed with strong security mechanisms. As a result, RPL-based IoT networks remain vulnerable to routing attacks such as sinkhole attacks and selective forwarding, which can significantly degrade network reliability and compromise data integrity. These issues are particularly critical in applications such as disaster management, healthcare monitoring, and smart infrastructure, where trustworthy data delivery is essential.

My individual component of the overall framework is the design and implementation of a trust and security-aware RPL routing mechanism. The aim is to enhance routing decisions by integrating trust and security factors into the RPL objective function, while maintaining the lightweight nature required for resource-constrained IoT devices.

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1.1 Purpose

The purpose of this Preliminary Progress Review (PPR) document is to present the current progress, methodology, and direction of my individual research component to the project supervisor and evaluation panel. This document highlights the work completed so far, explains the planned approach, and demonstrates that the research is progressing in alignment with the approved proposal and project timeline.

1.2 Abbreviations

Acronym	Description
IoT	Internet of Things
RPL	Routing Protocol for Low-Power and Lossy Networks
LLN	Low-Power and Lossy Network
OF	Objective Function
MRHOF	Minimum Rank with Hysteresis Objective Function
ETX	Expected Transmission Count
PDR	Packet Delivery Ratio

1.3 Overview

This document provides an overview of relevant literature, defines the research problem within my scope, outlines the research objectives and methodology, describes progress achieved so far, and presents the planned schedule and expected outcomes.

2.0 Literature Review

RPL has been extensively studied as the default routing protocol for IoT and LLN environments due to its lightweight operation and scalability. Standard objective functions such as OF0 and MRHOF focus on metrics like hop count and ETX to optimize routing performance. However, these objective functions do not consider node trustworthiness or malicious behavior, making RPL vulnerable to routing-based attacks.

Previous research has proposed trust-based and security-aware extensions to RPL, integrating factors such as historical behavior, residual energy, and reputation into routing decisions. While these approaches improve security, many introduce additional computational complexity, increased control overhead, or rigid parent selection mechanisms that reduce routing stability. This highlights the need for a balanced solution that enhances security without compromising performance.

2.1 Identification and Significance of the Problem

The main challenge identified in existing work is the trade-off between security and efficiency. Many trust-aware RPL mechanisms improve resilience against attacks but are unsuitable for highly constrained IoT devices due to increased overhead. Furthermore, static or inflexible parent selection strategies reduce adaptability under dynamic network conditions. Addressing these issues is essential for deploying secure and reliable IoT networks in real-world scenarios.

3.0 Research Question

- How can trust and security factors be integrated into the RPL objective function without introducing significant computational and energy overhead?
- Can a dynamic parent selection mechanism improve routing stability and resilience against routing attacks in RPL-based IoT networks?
- How does the proposed trust-aware routing mechanism compare with standard RPL in terms of packet delivery ratio, latency, and energy efficiency?

4.0 Research Objectives

4.1 Main Objective

To design and implement a trust and security-aware RPL routing mechanism that improves resilience against routing attacks while maintaining essential performance metrics in resource-constrained IoT environments.

4.2 Sub Objectives

- To modify the RPL objective function by incorporating trust and security-related factors.
- To design a dynamic parent selection mechanism that adapts to changes in trust conditions.
- To evaluate the proposed mechanism against standard RPL using performance metrics such as packet delivery ratio, latency, and energy efficiency.

5.0 METHODOLOGY

5.1 System diagram

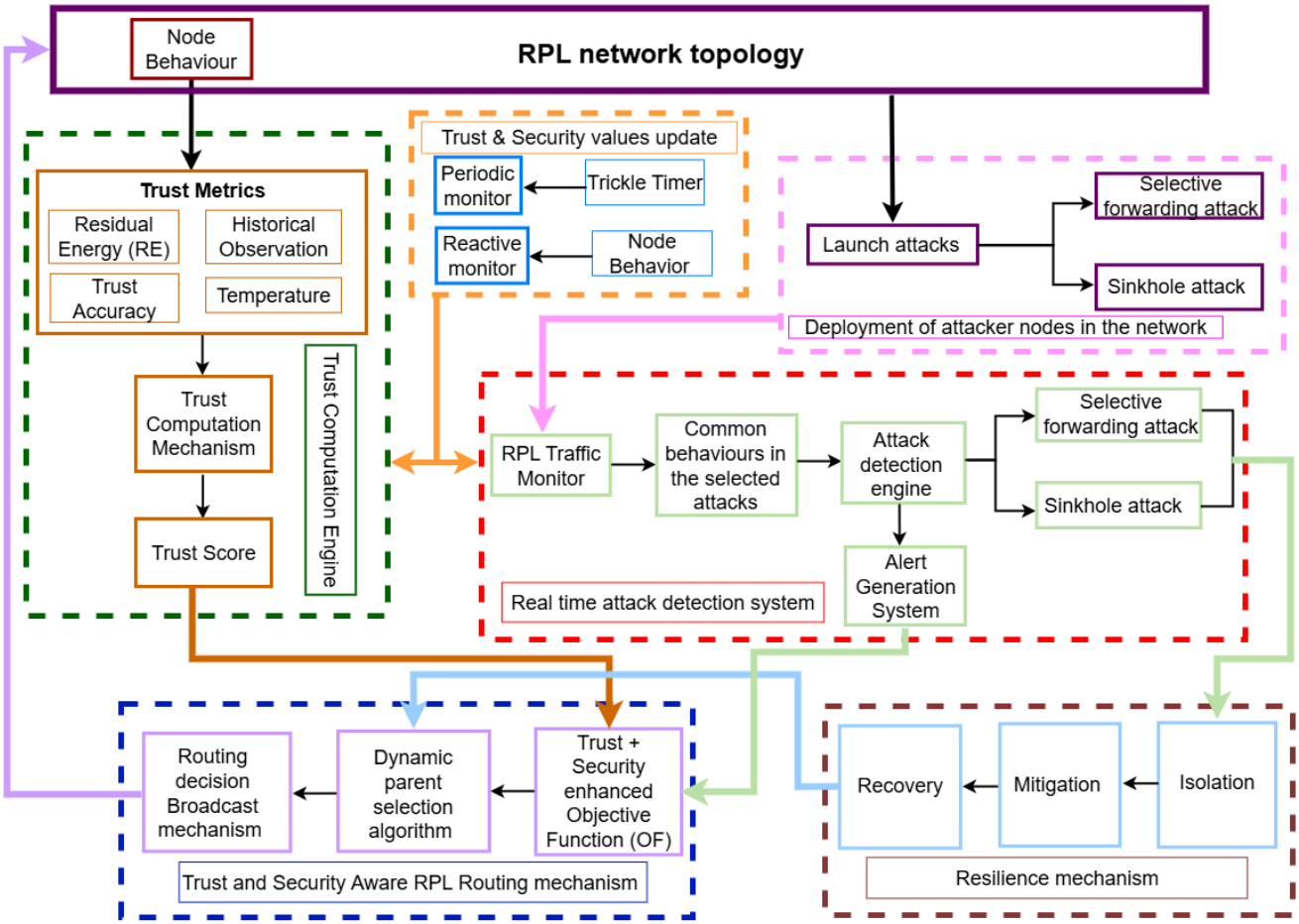


Figure 1: Lightweight Trust Management Framework for RPL-based IoT Networks

Figure 1 will present the proposed RPL network topology and the four main modules of the framework, the Trust Computation Engine, the RPL Traffic Monitor and Attack Detection Engine, the Trust & Security–Aware RPL Routing mechanism, and the Resilience Mechanism (isolation, mitigation, recovery). The data flow will begin with node behavior observation, each sensor node will monitor packet-forwarding behavior, link quality, residual energy and local indicators such as temperature. These raw observations will be input to the Trust Computation Engine, which

periodically and reactively computes a compact trust score for each neighbor using lightweight aggregation. The trust score output will then be pushed to the routing module. Simultaneously, the RPL Traffic Monitor will analyze control and data-plane flows to extract behavioral signatures, which will be fed into the Attack Detection Engine. This engine will raise alerts when suspicious patterns, such as selective forwarding or sinkhole behaviors, are detected. The routing module will then blend the trust scores, security alerts, and classical RPL metrics (ETX, hop count) into a composite "enhanced objective function" to select or re-rank parents. When a parent change is triggered, the routing module will issue secure route broadcasts so that neighboring nodes can update their local views. If the Attack Detection Engine raises a high-severity alert, the resilience module will isolate the node and trigger mitigation and recovery procedures. This layered structure will ensure that routing decisions remain adaptive, resilient, and lightweight, meeting the needs of resource-constrained IoT devices [1]–[3].

5.2 My Research Component Methodology

The core of this work will be the design and deployment of a modified RPL objective function and an adaptive parent selection policy that together will ensure secure routing decisions with minimal overhead. The modified objective function will be formulated so that a node's composite rank is a bounded combination of a traditional metric (such as ETX or base rank), a normalized trust term from the Trust Engine, and a security factor derived from detection alerts.

The trust term will be computed as a short-decay score that captures recent forwarding behavior and stability, together with one or two device health indicators (e.g., residual energy). The security factor will be event-driven and conservative, raising a parent's effective cost after a confirmed alert. Occasional false alarms will not immediately disrupt topology stability. To avoid oscillation, the parent selection policy will incorporate hysteresis windows and guard timers: small trust drops will not trigger immediate parent switching, whereas confirmed malicious behavior or repeated failures will force expedited re-selection.

All design decisions (including which metrics to combine, normalization ranges, and threshold values) will be parameterized so that they can be tuned during experiments. The implementation will modify the RPL module in Contiki-NG by extending neighbor tables to include trust/security fields and by adding the composite ranking step to the parent-ranking routine. The modified DIO/DAO behavior for broadcasting trust-aware routing decisions will be designed to remain backward-compatible, and lightweight message authentication will be integrated to protect against manipulation of trust-related fields without imposing heavy cryptographic costs.

5.3 Tools, Software, and Environment

The implementation will be carried out using Contiki-NG, an open-source operating system widely used for IoT simulations due to its accurate modeling of RPL behavior and support for low-power lossy networks. Its Cooja simulator provides node-level energy models, traffic visualization, and customizable attack scenarios, making it highly suitable for this research [24][8]. Experiments will be conducted on personal PCs with at least 16 GB RAM and 512 SSD, allowing multiple simulation runs in parallel. Development will be in C for the protocol stack modifications and Python for simulation orchestration, data collection, and statistical analysis. Code versioning will be managed using GitHub, while Microsoft Teams and WhatsApp will support group collaboration.

5.4 Data needs and collection strategy

All primary data will be produced by the Cooja/Contiki simulations: per-packet traces, per-node energy counters (simulated), parent selection logs, DIO/DAO counts, trust values over time and attack alert traces. To create realistic attack patterns and validate detection thresholds, public datasets such as IoT-23 and UNSW-NB15 will be used as references for traffic patterns and malicious behavior synthesis, while Intel Lab and SmartSantander traces are used to select sensible traffic models and sensor sampling intervals [14][9]. Each scenario will be repeated multiple times (we will

use at least 30 runs per configuration) to provide statistical confidence; logs are exported as CSV files and processed in Python to compute means, standard deviations and confidence intervals.

5.5 Experimental design and evaluation criteria

Experiments will compare the proposed mechanism against standard RPL (OF0 and MRHOF) under four primary scenarios: normal (no attack), with attack (sinkhole or selective forwarding), with Trust issues, and mixed attacks + Trust. Network sizes will be scaled (for example, 20, 50 and 100 nodes) to study scalability. Metrics include packet delivery ratio, end-to-end latency, control overhead (DIO/DAO counts), parent-switch frequency (stability), estimated energy overhead, detection rate and false positive rate, and time-to-mitigation. Success will be measured by the system's ability to maintain delivery and latency performance within acceptable deviation from baseline RPL, while significantly improving resilience against attacks. Statistical tests will be conducted to confirm the validity of results [10] [11] [14] [17].

5.6 Tasks, sub-tasks and milestones

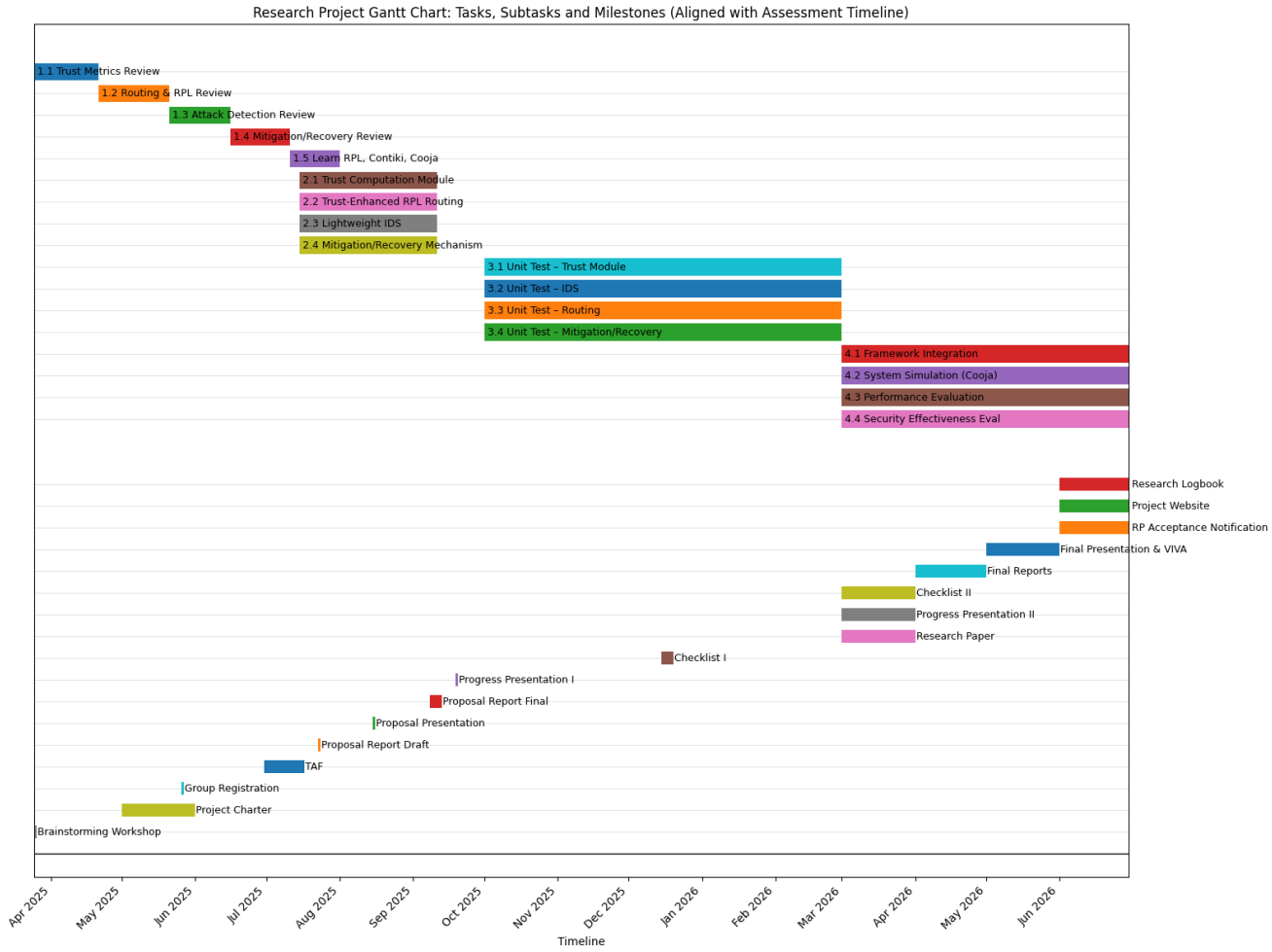


Figure 2: Overall Research Timeline (Gantt Chart)

The first phase focuses on literature review and requirements definition, synthesizing existing work on RPL security, trust metrics, and intrusion detection systems to establish the foundation for design. This phase also defines evaluation criteria such as packet delivery ratio, energy efficiency, and attack resilience.

The second phase involves algorithm and model design. Here, the mathematical formulation of the composite objective function is developed, and the parent selection algorithm is defined with stability

mechanisms such as guard timers. The design phase also specifies secure broadcast formats and lightweight authentication to protect against manipulation of trust-related control messages [7].

The third phase is the implementation, where the enhanced objective function and parent selection policies are integrated into Contiki-NG. The implementation also includes secure broadcast handling, data logging mechanisms for simulation, and APIs to integrate the trust and detection modules.

The fourth phase is framework integration, where modules from group members such as the Trust Engine and Attack Detection System—are combined into a single system. This requires defining message formats, synchronization of update intervals, and collaborative debugging.

The final phase is evaluation, where the complete system is tested under multiple IoT scenarios with varying network sizes and attack intensities. Metrics such as delivery ratio, latency, control overhead, and resilience will be measured, with statistical analysis (t-tests or non-parametric methods) applied to validate significance.

The literature review, system design, and initial proposal milestones conclude with the Proposal Report Draft (August 31, 2025) and Proposal Presentation (September 8–12, 2025). Implementation and testing phases coincide with Progress Presentation I in October 2025, while integration and extended evaluation support Progress Presentation II in March 2026. The final results, reports, and viva are scheduled between April and May 2026, ensuring that the research outputs align with academic assessment deadlines.

5.7 Anticipated results

It is expected that the proposed mechanism will significantly reduce the impact of routing attacks while preserving key performance metrics. Specifically, improvements are anticipated in packet delivery ratio, network stability, and resilience to malicious behavior, with minimal additional energy or delay overhead. The integration of trust and security factors into the objective function is expected to result in more adaptive parent selection, preventing prolonged reliance on compromised nodes.

By addressing both trustworthiness and security at the routing layer, this research aims to provide a lightweight, practical, and scalable enhancement to RPL. The findings will not only demonstrate a balanced trade-off between performance and security but also contribute design insights for future IoT routing protocols.

6.0 Current Progress

Up to this stage, the following progress has been achieved for my research component:

- Successfully set up the Contiki-NG and Cooja simulation environment.
- Compiled and executed baseline RPL applications using MRHOF to verify correct routing behavior.
- Implemented a standalone metrics module to handle trust, security factors, and link quality metrics.
- Developed the initial scaffold of the trust and security-aware objective function using RPL Lite APIs.
- Implemented parent ranking and selection logic and analyzed parent selection behavior through simulation logs.

These steps correspond to the foundational stages required for integrating trust-aware routing logic and provide a stable base for further enhancements such as hysteresis tuning and attack mitigation.

7.0 Anticipated Benefits

7.1 Benefits to Users

- Improved reliability and trustworthiness of IoT routing decisions.
- Enhanced resistance to routing attacks without heavy computational overhead.
- More stable routing paths in dynamic network conditions.

7.2 Contribution to the Body of Knowledge

This research contributes a lightweight and practical trust-aware routing approach for RPL-based IoT networks, addressing the limitations of existing security-focused solutions.

8.0 Scope and Expected Research Outcome

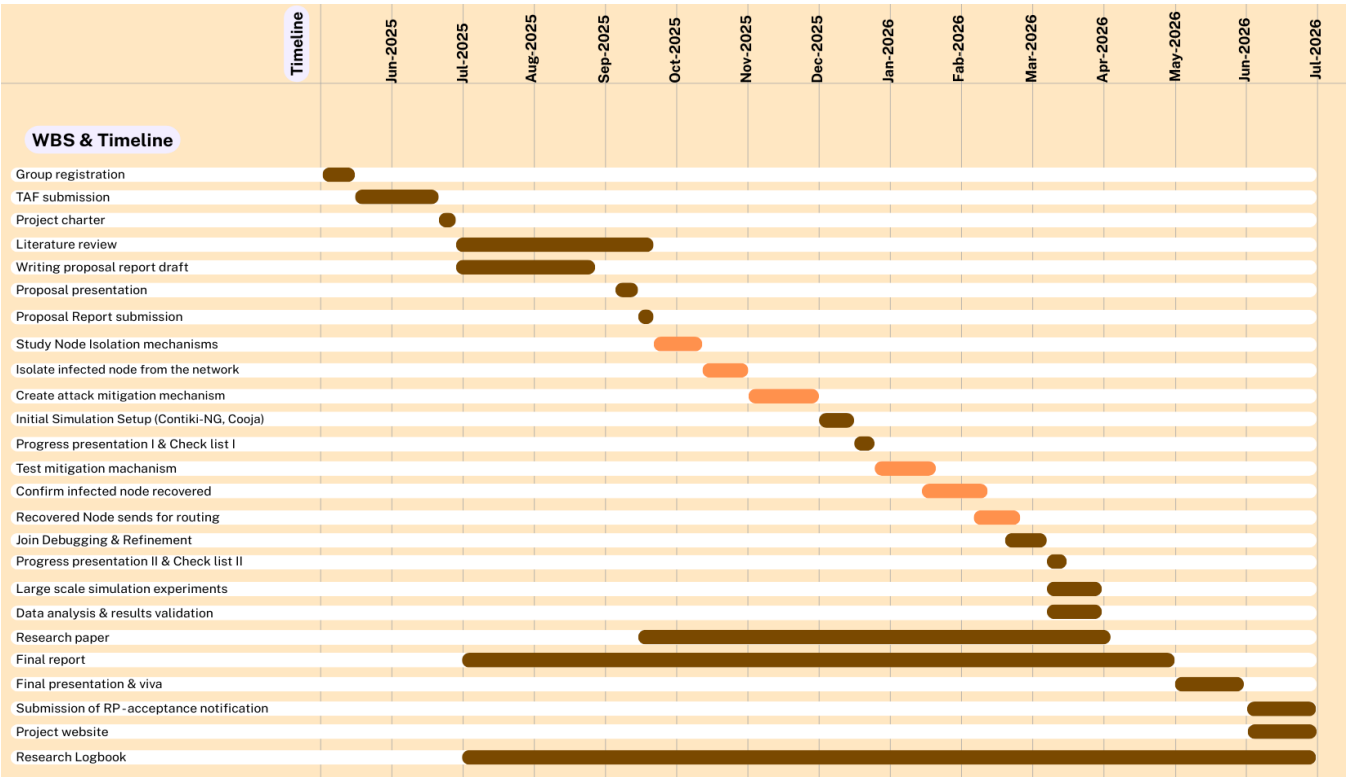
The scope of this research is limited to the design, implementation, and evaluation of a trust and security-aware RPL routing mechanism within a simulated IoT environment.

The expected outcomes include a modified RPL objective function, a dynamic parent selection algorithm, and experimental results demonstrating improved resilience against routing attacks compared to standard RPL.

9.0 Research Constraints

- Limited simulation time and computational resources.
- Dependence on simulated attack scenarios rather than real-world deployments.
- Constraints imposed by the lightweight nature of IoT devices.

10.0 Project Plan



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